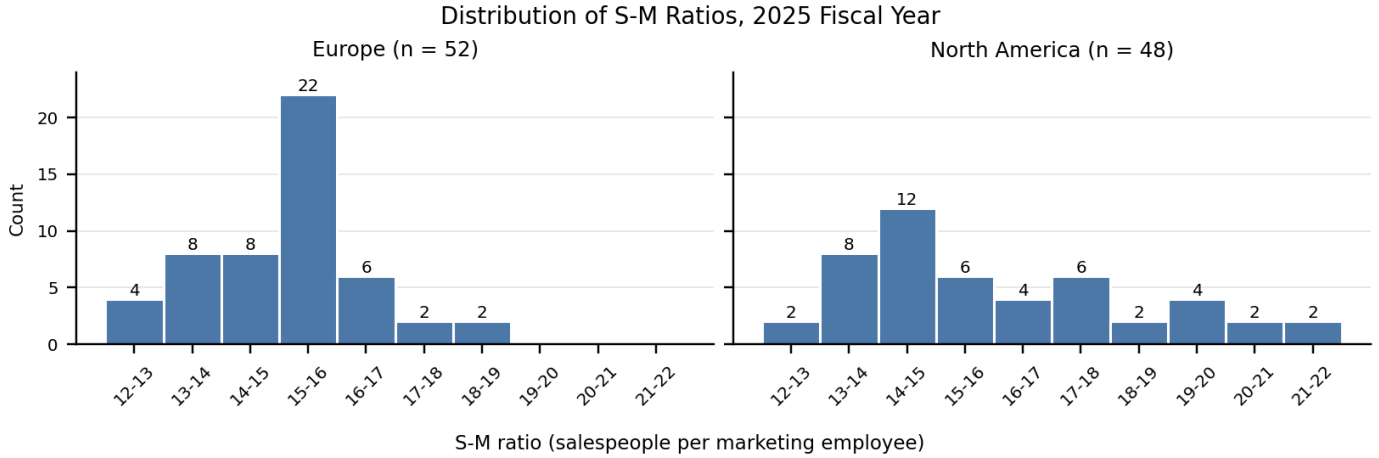
SOCS Comparison Practice: S-M Ratios in Europe and North America

Show all your work. Indicate clearly the methods you use, because you will be graded on the correctness of your methods as well as on the accuracy of your results and explanation.

A multinational corporation calculates the **sales-to-marketing personnel ratio (S-M ratio)**, defined as the **number of salespeople per marketing employee**, for each of its branches. The histograms show the S-M ratios for the **52 branches in Europe** and the **48 branches in North America** during the 2025 fiscal year.



Region	n	Mean	SD	Min	Q1	Median	Q3	Max
Europe	52	15.12	1.37	12.3	14.2	15.3	15.9	18.3
North America	48	16.06	2.36	12.6	14.4	15.3	17.1	21.8

Summary statistics are calculated from the raw data. **Extreme values:** Europe — two smallest 12.3, 12.5; two largest 18.1, 18.3. North America — two smallest 12.6, 12.9; three largest 20.6, 21.3, 21.8.

Task (20 points): Compare the distributions of S-M ratios for the branches in Europe and North America. Address **shape, outliers, center, and spread**, justify your answers with appropriate calculations, and **interpret your results in context**.

Comparison Checklist

Shape: Describe the shape of **each** distribution (peaks; symmetric or skewed). State how the two shapes **differ**. Give evidence.

Outliers: Use the 1.5 × IQR rule to find the fences for **each** region. Check the extreme values. Which region has outliers, and which values?

Center: Choose the better measure and explain why. Report it for **both** regions and compare. Do the means tell the same story? Why or why not?

Spread: Choose the better measure and explain why. Report it for **both** regions and compare. Describe the middle 50% in each region.

Context: Name both groups. Use units. Explain what the comparison means for the branches.

Toolbox

IQR = Q3 − Q1
Upper fence = Q3 + 1.5 × IQR
Lower fence = Q1 − 1.5 × IQR
An outlier is a value **beyond** a fence. With skewness or outliers, compare **medians** and **IQRs**.

Phrase Bank

Both distributions are …
Europe is …, while North America is …
Unlike Europe, North America …
The medians are the same (… vs. …).
… is more variable than …
The IQR for … is larger than …
… has … outliers, but … has none.